Final Exam Prepare - 30/9 exam

Paper - 2025 OCT & 2025 MAY, 全部题目一起来

2025 OCT

Q1- 全部固定format的calculate

a) Perform the following conversions. Show the working steps clearly. If the operation(s) is invalid, explain the reason.

(i) 7119_7 to base-10 number (2 marks)

(ii) 1101110.11_2 to base-16 number (3 marks)

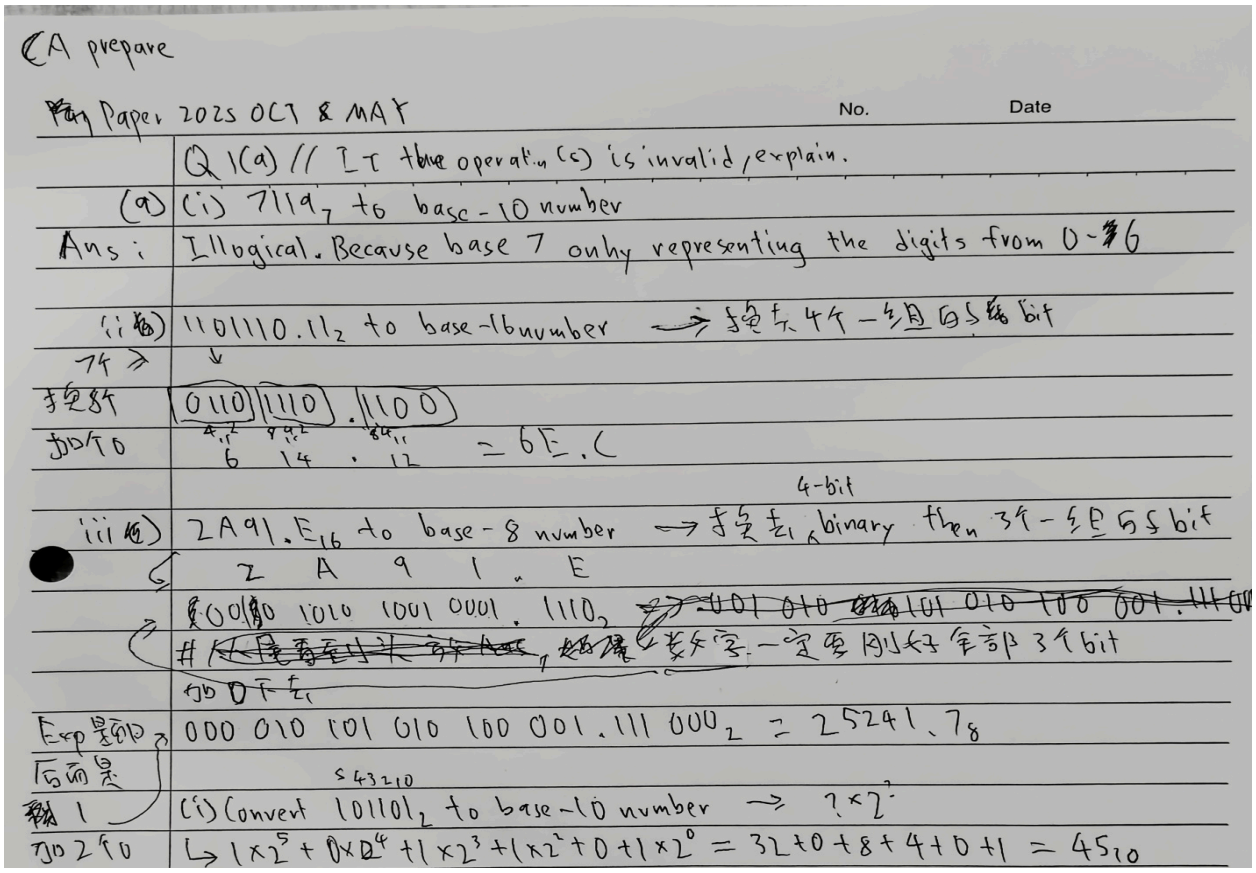
(iii) $2A91.E_{16}$ to base-8 number (4 marks)

Invalid Explain Format:

Illogical.

Because, base ??(看题目是什么数字) only representing the digits from ?? (Exp, 16 = 0-9, A-F)

a) Ans:

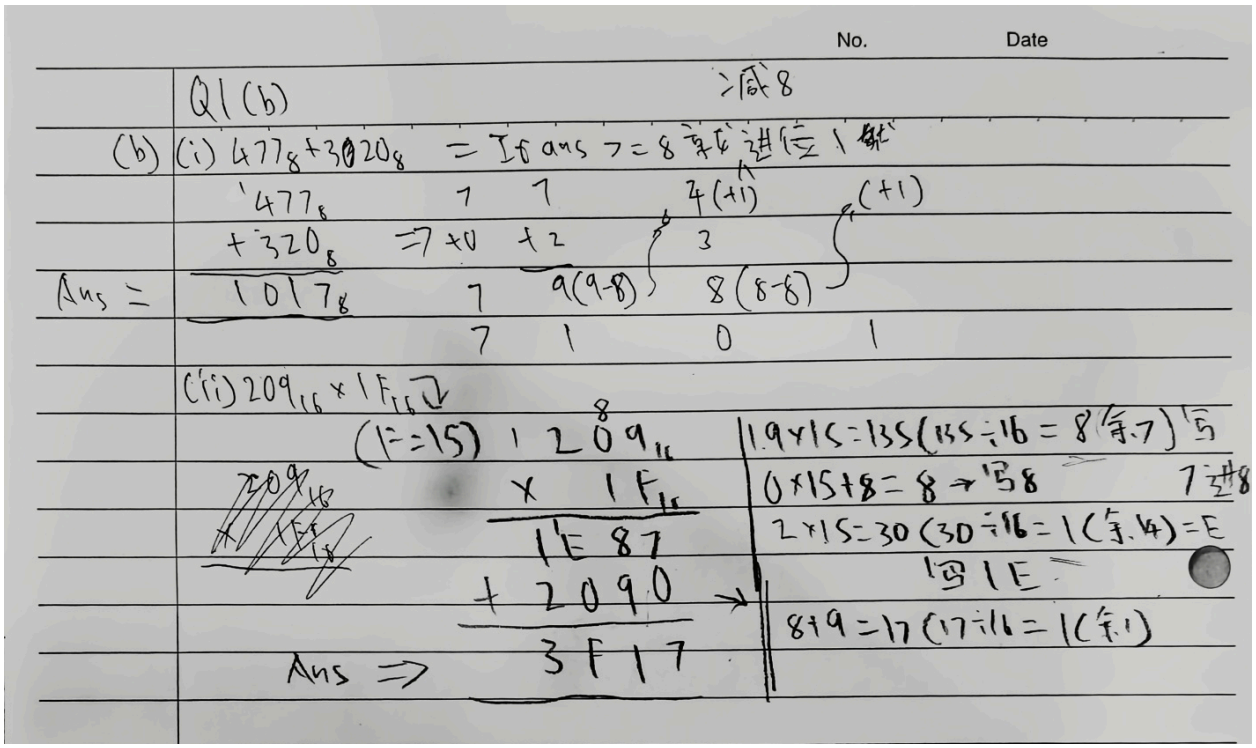


b) Perform the following arithmetic operations. Show the working steps clearly.

(i) $477_8 + 320_8$ (3 marks)

(ii) $209_{16} \times 1F_{16}$ (3 marks)

b) Ans:



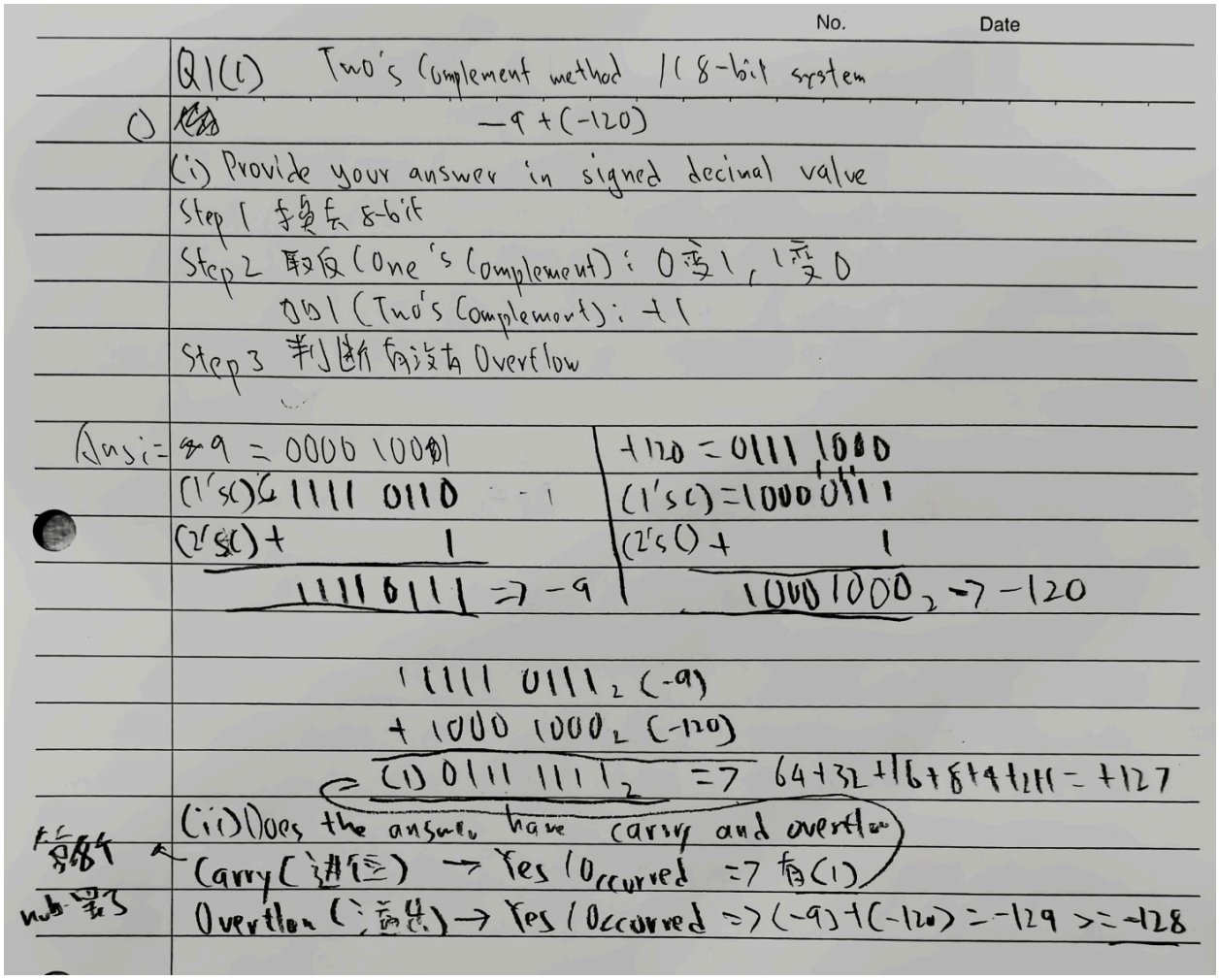
c) Assuming an 8-bit system is used, solve the following operation using the Two's Complement method.

$$-9 + (-120)$$

(i) Provide your answer in signed decimal value. (4 marks)

(ii) Does the answer have carry and overflow? (1 mark)

c) Ans:



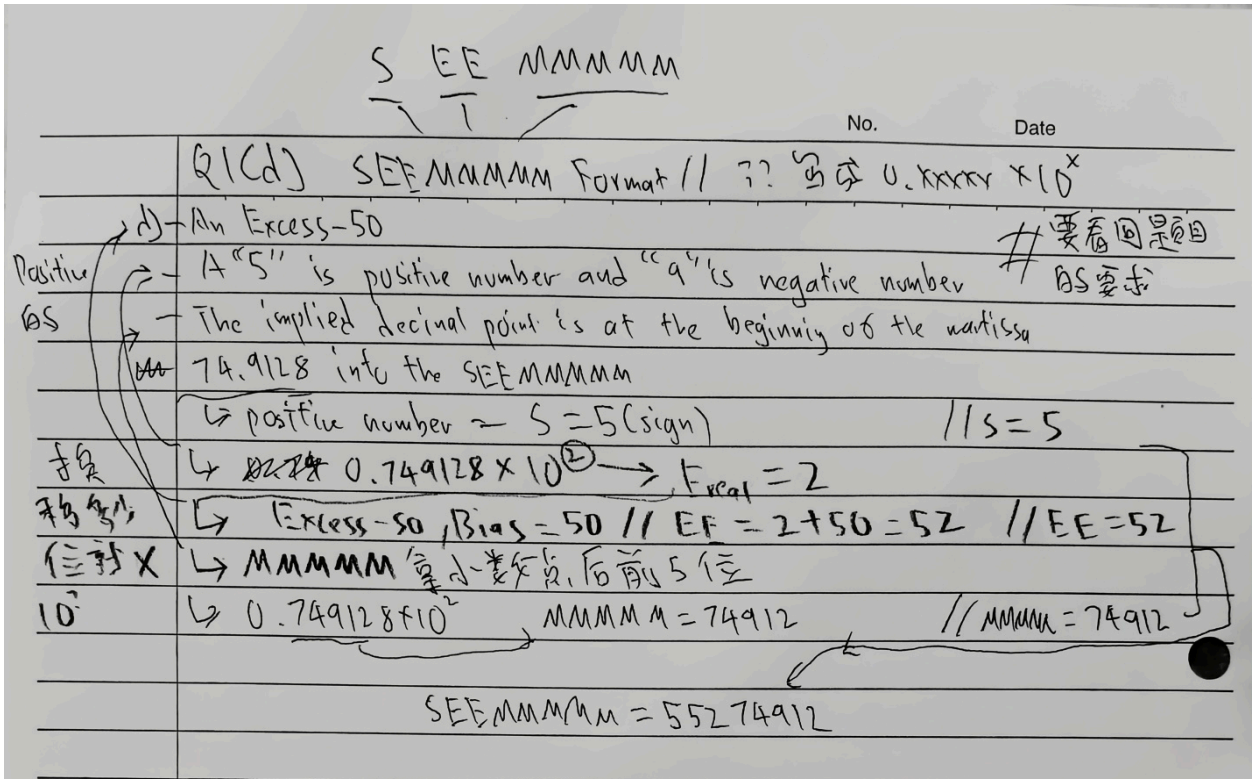
d) The SEEMMMMM format is used to represent floating-point numbers, where 'S' denotes the sign, 'EE' represents the exponent, and 'MMMMMMM' represents the mantissa.

Given:

- An Excess-52 notation is applied.
- The mantissa has an implied decimal point immediately before it.
- A "3" represents a positive number, while a "6" represents a negative number.

Convert -45.6789 to the SEEMMMMM format. (5 marks)

Ans:



2025 MAY

Question 1

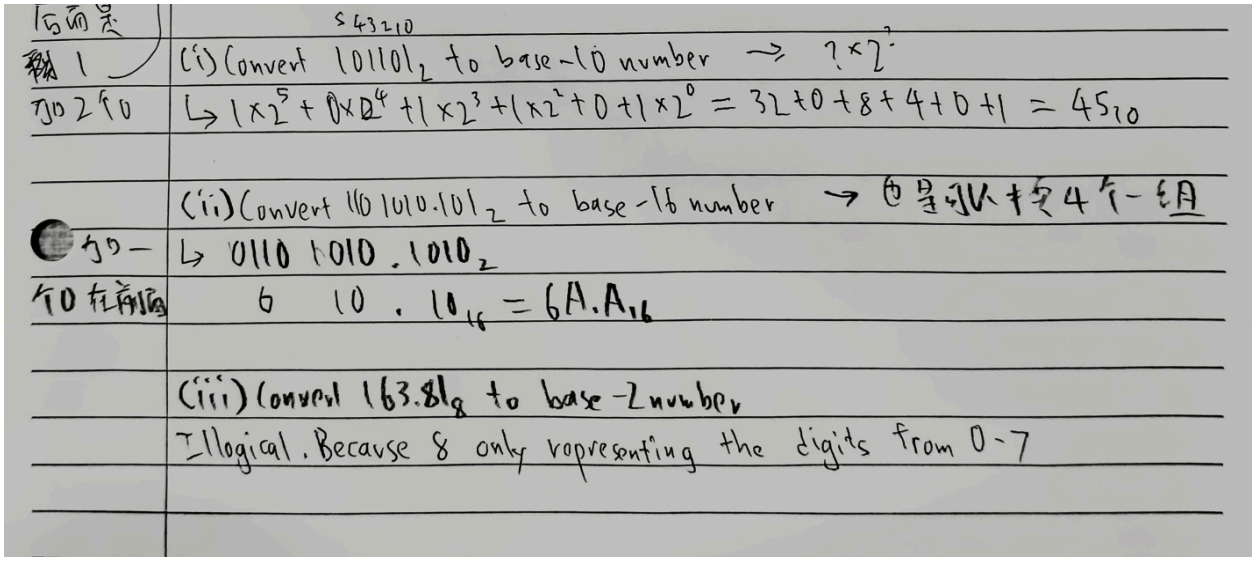
a) Perform the following conversions. Show all working steps clearly. If the operation(s) is invalid, provide an explanation.

(i) Convert 101101_2 to base-10 number. (2 marks)

(ii) Convert 1101010.101_2 to base-16 number. (3 marks)

(iii) Convert 163.81_8 to base-2 number. (3 marks)

a) Ans:

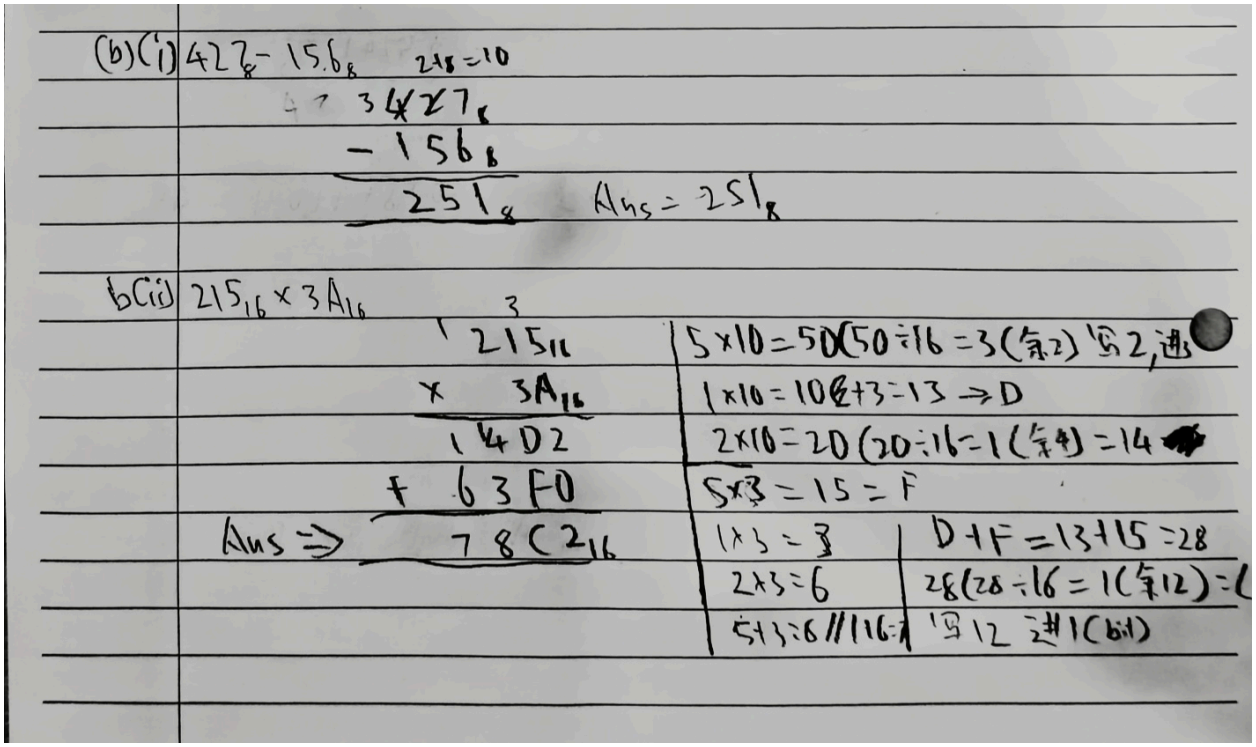


b) Solve the following operations. Show all working steps clearly.

(i) $427_8 - 156_8$. (3 marks)

(ii) $215_{16} \times 3A_{16}$. (3 marks)

b) Ans:



c) Assume an 8-bit system is applied. Solve the following operation using the Two's Complement method.

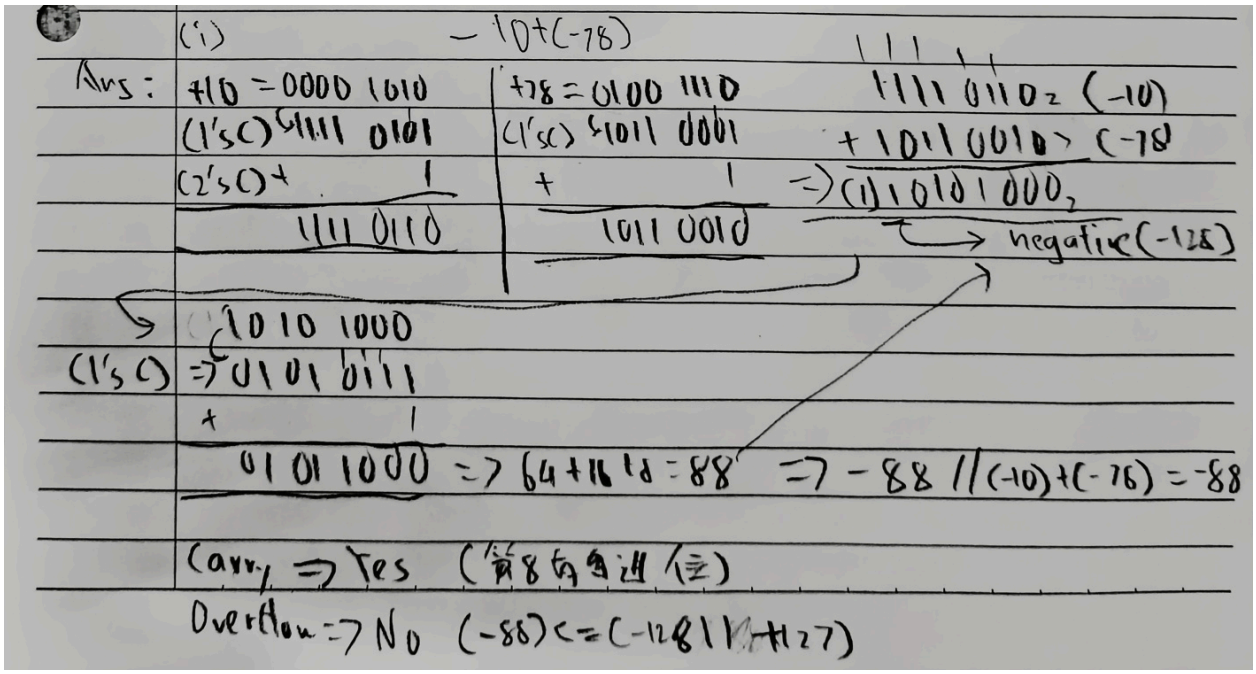
-10 + (-78)

(i) Provide the result in signed decimal form. (4 marks)

(ii) Identify whether a carry occur. (1 mark)

(iii) Identify whether an overflow occur. (1 mark)

c) Ans:



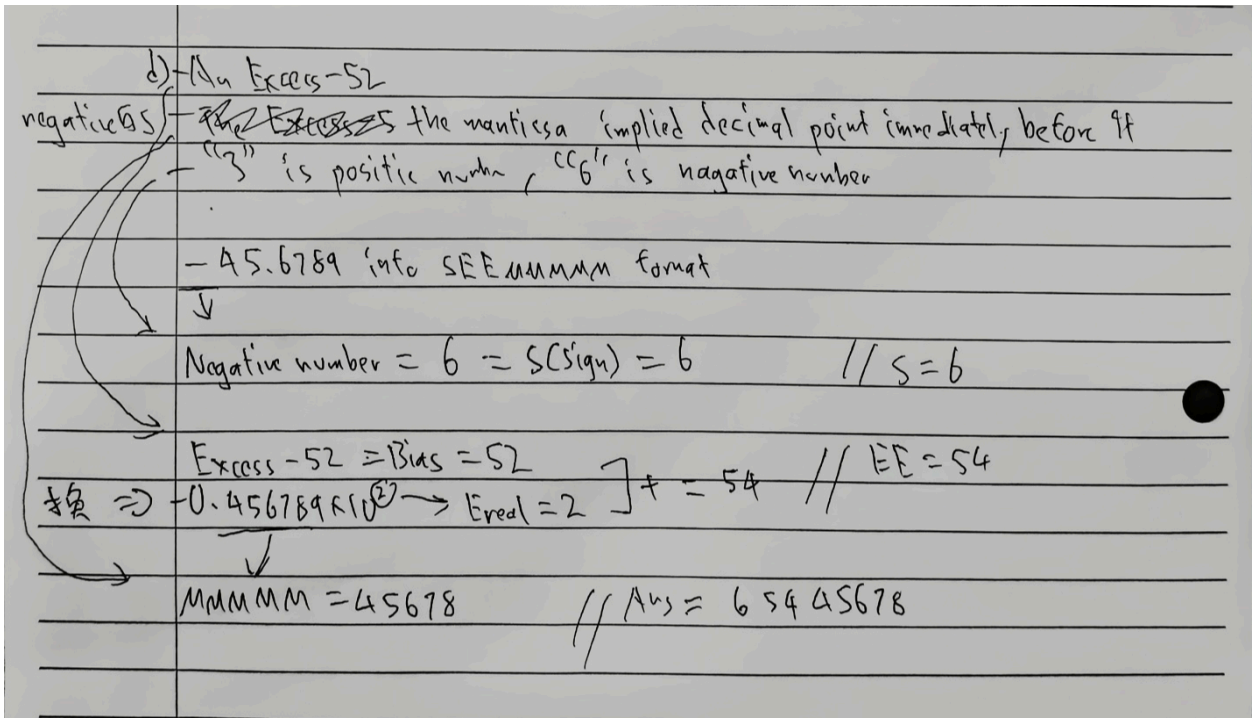
d) The SEEMMMMM format is used to represent floating-point numbers, where 'S' denotes the sign, 'EE' represents the exponent, and 'MMMMMMM' represents the mantissa.

Given:

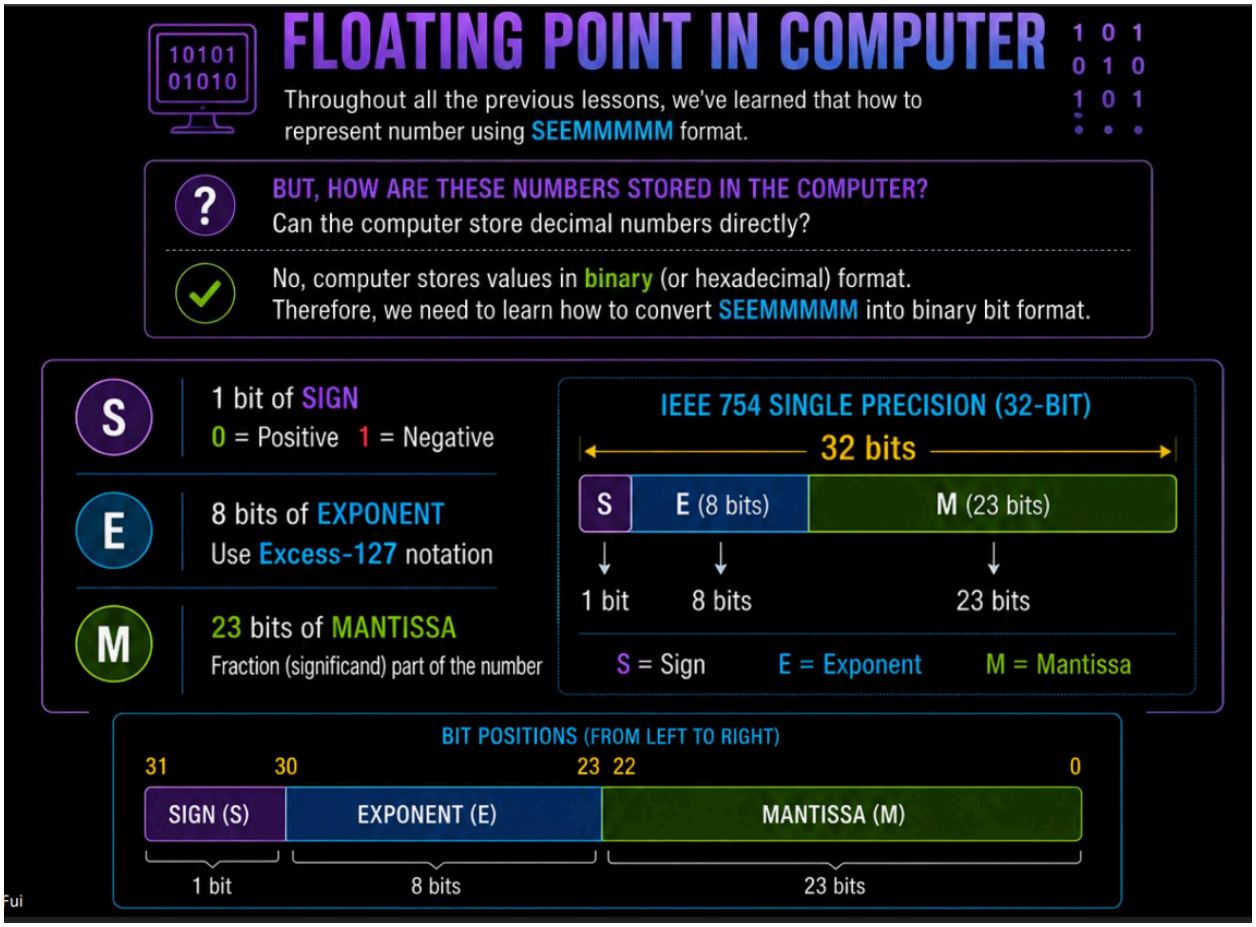
- An Excess-52 notation is applied.
- The mantissa has an implied decimal point immediately before it.
- A "3" represents a positive number, while a "6" represents a negative number.

Convert -45.6789 to the SEEMMMMM format. (5 marks)

Ans:



Extra calculation（In Q1）



转换步骤（6 Steps）

Step 1: Convert to Binary

方法：
整数部分：不断除以 2，取余数，倒序读。
小数部分：不断乘以 2，取整数部分，顺序读。

Example:
$13.25_{10} = 1101.01_{2}$

Step 2: Normalize
把二进制数写成 1.xxxx × 2^n 的形式。

方法:
小数点移到第一个 1 的后面。
数小数点移动了几位, 就是 exponent。

Example:
$1101.01_2 = 1.10101_2 \times 2^3$
↑
Exponent = 3

Step 3: Determine Sign (S)
正数 → S = 0
负数 → S = 1

Example:
13.25_{10} is positive → S = 0

Step 4: Determine Exponent (E)
Format:
Stored Exponent = Actual Exponent + Bias (127)

Example:
Actual Exponent = 3
Stored Exponent = $3 + 127 = 130_{10}$
$130_{10} = 10000010_2$

Step 5: Determine Mantissa (M)
方法：
从 normalized form 拿掉 leading 1。
剩下的 fractional part 就是 mantissa。
补 0 到 23 bits。

Example:
1.10101_2 → 拿掉 leading 1 → 10101
Pad with zeros to 23 bits:
10101000000000000000000

Step 6: Assemble the 32-bit IEEE 754 Format
把 S + E + M 拼起来。

Example:
S = 0
E = 10000010
M = 10101000000000000000000

32-bit Pattern:
0 10000010 10101000000000000000000

Paper 2025 JAN

d) Given a decimal number -89.75_{10}, show how this notation can be represented in the IEEE754 single precision notation using Excess-127 notation. You are required to show your working steps. (5 marks)

d) Ans:

Extra

No. Date

Paper 2025 JAN

S E M M M M M M

1 8 23 = 32 bit

d) number -89.75 // IEEE 754 // Excess-127 notation

Step 1 Convert number to binary

-89.75

↳ 1011001.11_2

89 2|89 - 1
2|44 - 0
2|22 - 0
2|11 - 1
2|5 - 1
2|2 - 0
2|1 - 1

1011001_2

0.75 → 0.11

↳ 0.75 × 2 = 1.50 (拿1)

↳ 0.50 × 2 = 1.00 (拿1)

Step 2 Normalize 把 binary 換去 $1.xxx \times 2^?$

$1011001.11 \rightarrow 1.01100111_2 \times 2^6$ 有 Exponent = 6

↳ 移6次

Step 3 Determin Sign

Positive number = S → 0

Negative number = S → 1 → (-89.75)

↳ S = 1

Step 4 Determine Exponent (E)

Stored Exponent = Actual Exponent + Bias(127) / (Excess-127 notation)

6 + 127

= 133_{10} convert 去 binary

↳ 10000101 8 bit

Step 5 Determine Mantissa

1.01100111, 拿掉1 = 01100111 + (15个0) = 01100111000000000000000

Explain, Mantissa 係 23 bit

So 23 - 8 = 15 bit, 直接加15个0

Step 6 Gp 起來拿去 32 bit

Ans: 1 10000101 01100111000000000000000

S E M ※Ans 是連起來的

Q2 - a)&b)固定calculate, c) d) e) f) Chap3的theory

2025 OCT

a) The colour value of a pixel is represented using 3 bytes, with each byte corresponding to one colour component: Red, Green, and Blue.

Given a 1-pixel image stored in memory starting at address 0x6009, the RGB values for each pixel are as follows:

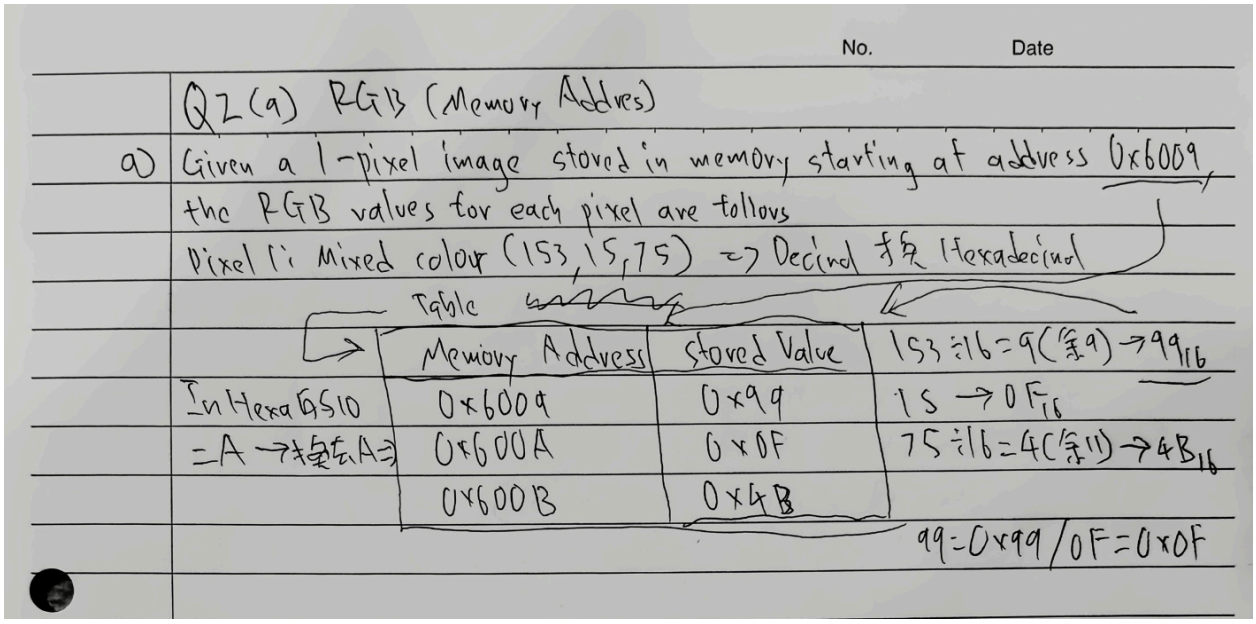
Pixel 1: Mixed colour (153, 15, 75)

Using hexadecimal format, show the value stored at each memory address for this pixel in a table:

Memory Address	Stored Value

(3 marks)

a) Ans:

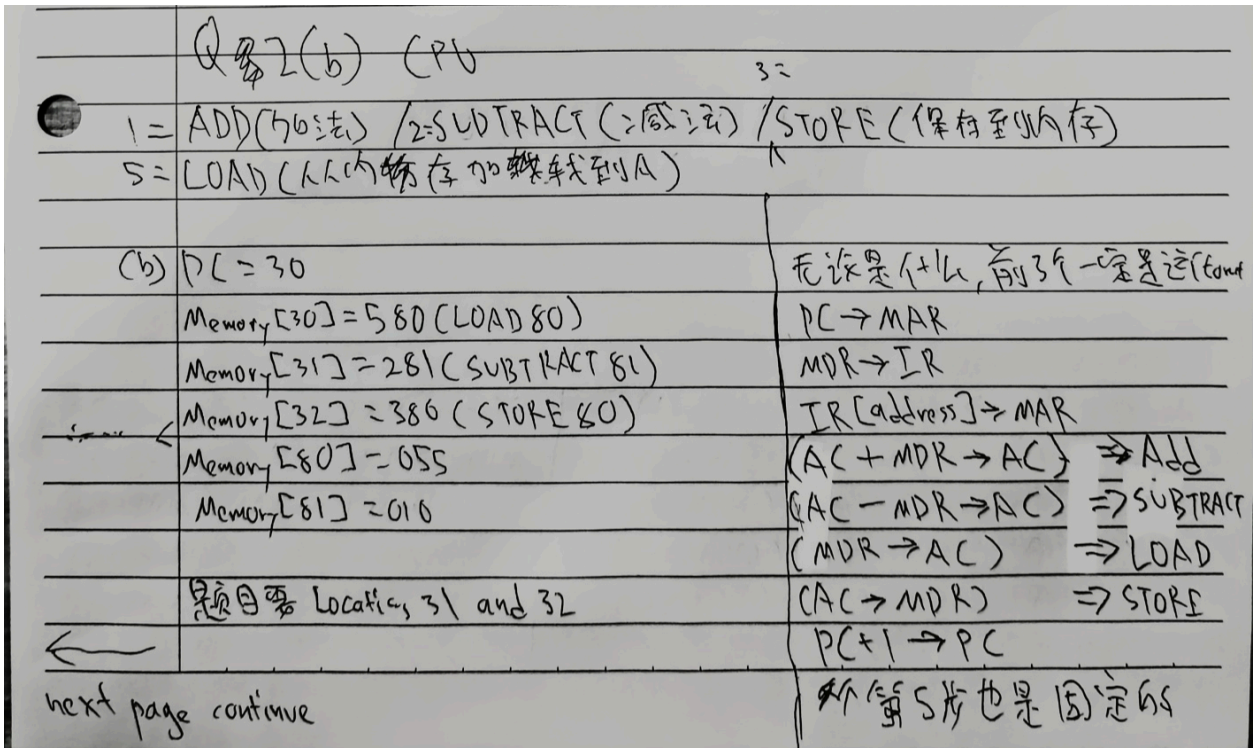


b) Assuming that the Little Man Computer (LMC) model is applied. Suppose that the following instructions are found at the given location in memory:

Program counter: 30
Value in memory location 30: 580 (LOAD)
Value in memory location 31: 281 (SUBTRACT)
Value in memory location 32: 380 (STORE)
:
Value in memory location 80: 055
Value in memory location 81: 010

Show the changes of contents for Instruction Registers (IR), Program Counter (PC), Memory Address Register (MAR), Memory Data Register (MDR), and Accumulator (A) during the execution of instructions at locations 31 and 32. (10 marks)

b) Ans:



No. Date

(Q2(b)(cont.)

Memory[31] = 281 (SUB 81)

Memory[32] = 380 (STORE 80)

↳ 从 memory[80] = 055 拿进来 A = 055, 因为 [30] = 580 (LOAD)

把[80]存进来

A = 055

Location 31

Ans:

↳ 现在 A = 055 ⇒

↳ PC → MAR (MAR = 31)

MDR → IR (IR = 281)

IR[address] → MAR (MAR = 81)

A - MDR → A (A = 055) - (MDR = 010) = (A = 045)

从[81]拿的(010)

PC + 1 → PC (PC = 32)

Location 32

Ans.

Memory[32] = 380 (STORE 80)

↳ PC → MAR (MAR = 32)

MDR → IR (IR = 380)

IR[address] → MAR (MAR = 80)

A → MDR (MDR = 045)

PC + 1 → PC (PC = 33)

2025 MAY

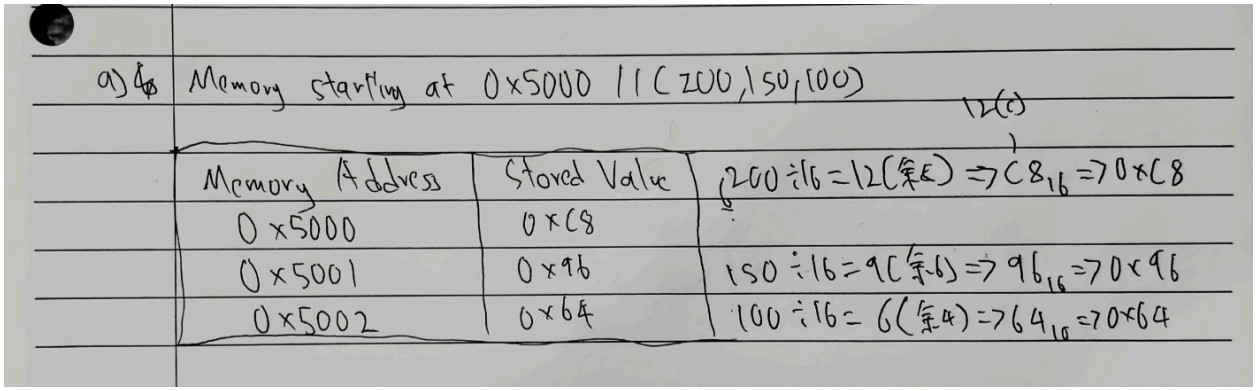
a) A 1-pixel image is stored in memory starting at address 0x5000, with the following RGB values:

Pixel 1: Mixed colour (200, 150, 100)

The colour value for the pixel is represented using 3 bytes, where each byte corresponds to one colour component: **Red**, **Green**, and **Blue**.

Using hexadecimal format, show the value stored at each memory address for this pixel in a table:

a) Ans:



a) Memory starting at 0x5000 ((200,150,100)

Memory Address	Stored Value
0x5000	0xC8
0x5001	0x96
0x5002	0x64

200 ÷ 16 = 12(餘8) ⇒ $C8_{16}$ ⇒ 0xC8

150 ÷ 16 = 9(餘6) ⇒ 96_{16} ⇒ 0x96

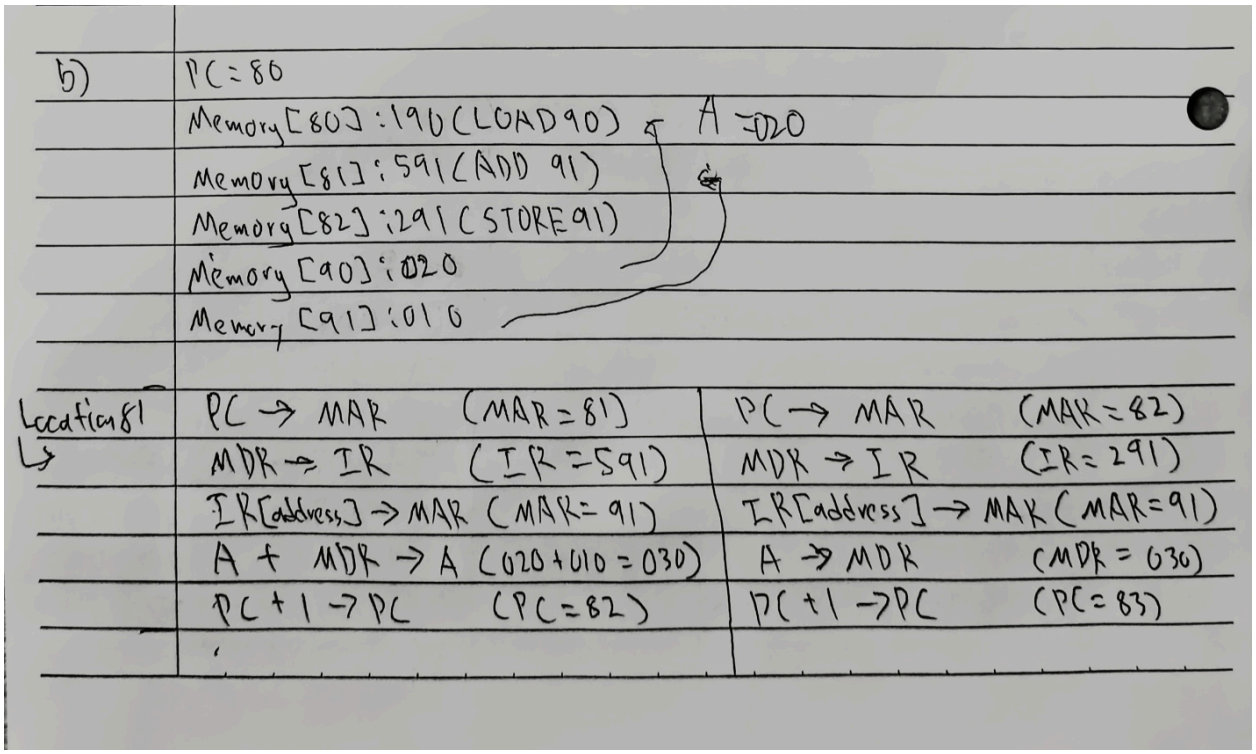
100 ÷ 16 = 6(餘4) ⇒ 64_{16} ⇒ 0x64

b) Using the Little Man Computer (LMC) model, consider the following instructions stored at the specified memory locations:

Program counter: 80
Value in memory location 80: 190 (LOAD)
Value in memory location 81: 591 (ADD)
Value in memory location 82: 291 (STORE)
:
Value in memory location 90: 020
Value in memory location 91: 010

Trace and show the changes in the contents of the *Instruction Registers (IR), Program Counter (PC), Memory Address Register (MAR), Memory Data Register (MDR), and Accumulator (A)* during the execution of instructions at locations 81 and 82. (10 marks)

b) Ans:



Theory part - 因为Q3也是chap 3的theory, 所以连通去Q3的theory部分

Components of CPU (ALU, CU, Interface Unit, register)

1. Role and Operations of the ALU (Page 11)

- It performs arithmetic calculations and Boolean logical calculations.

2. Role of the Control Unit (CU) (Page 11)

- Decode instructions.
- Controls instruction execution.

3. Interface Unit (Page 11)

- Moves instructions and data between the CPU and other devices

4. Types of Separate Buses (Page 31 - 35)

Data Bus

- Carries data being moved from one location to another.
- Bidirectional bus.

Address Bus

- Specifies the recipient of data on the bus.
- Unidirectional bus.

Control Bus

- Provides control for proper synchronization.
- Bidirectional bus.

5. Point-to-Point bus VS Multipoint Bus(Page 36-38)

	Point-to-Point Bus	Multipoint Bus
concepts	A bus that carries signals from a specific source to a specific destination.	A bus that connects several points together.
connection	One source to one destination	one source to multiple destination
example	Control unit to ALU	Ethernet

6. CISC VS RISC (Pages 41–49)

Concepts	
CISC	RISC
Large and complex instruction set	Small and simple instruction set
Variable-length instructions	Fixed-length instructions
Examples: Intel x86, Intel Core i9, AMD Ryzen	Examples: ARM, Apple M-Series, RISC-V

Characterized	
CISC	RISC
Few general-purpose registers.	32 general-purpose registers.
Many addressing techniques.	Limited addressing modes

Different	
CISC	RISC
Emphasis on hardware	Emphasis on software
Less registers	Uses more registers
More addressing modes	Fewer addressing modes

7. RISC-V Architecture (Page 51–59)
Open and free
- Free to use by anyone, for any application
Scalable and flexible architecture
- Supports multiple word sizes

Q3 - 大部分是Chap3 的 theory, 偶尔会出Chap 4的 calculate

Calculation part我丢在跟Q4那边一起，因为出的都是chap4的

Theory part (cont.)

8. IPO Version (1 Real Example)

Real-world example: Using a calculator.

1. Input:
   - The user enters data via the keyboard (e.g., typing "2 + 3").

2. Process:
   - The CPU processes the input according to the instructions (e.g., performs addition).

3. Output:
   - The result is produced and sent to an output device (e.g., the display shows "5").

4. Storage:
   - Data and information are stored for future use (e.g., saved to memory or history).

Input - Data and instructions entered into the computer.
Process - The CPU processes the input data according to instructions.
Output - The results of processing are produced and sent to output devices.
Storage - Data and information are stored for future use.

9. Components of Computer Systems (Page 5)

Hardware Component — Tangible physical parts of the system
Software Component — Intangible parts of the system

Data Component — Computer stores and processes data in binary
Communication Component — Communication channels and interface hardware

10. Register (Page 12)
- A special, high speed storage area within the CPU.
- Hold data temporary for storage, manipulation, and calculations.

11. Registers VS Memory (Page 12)

Registers	Memory
Small, very fast storage locations inside the CPU.	Larger, slower storage
Used for immediate data processing	Used for long-term data storage

12. Three Type of register (Page 13-14)
GENERAL PURPOSE REGISTERS
- Store data/result for arithmetic operations.

SPECIAL PURPOSE REGISTERS
- PC
- IR
- MAR
- MDR

FLAG REGISTERS
- 1 bit register
- CF, ZF, SF, OF, PF

13. The Operation of Memory (Page 17)
MAR - Holds the address in memory which is open for data access.

MDR - Store the data to be stored into memory in case of write.

14. Machine Cycle - four phase (page 20-21)
- FETCH: Retrieve an instruction from memory.
- DECODE: Translate the instruction into computer commands
- EXECUTE: Execute the computer commands.
- STORE: Send and write the results back in memory.

Q4 - 大部分是Chap4 的 theory, 偶尔会出Chap4的 calculate

Theory part

1. Types of Memory (Volatile vs. Non-Volatile) (Page 4)

	Volatile 易失性	Non-Volatile 非易失性
Concept	Hold data temporarily while CPU is processing	Hold data permanently
Storage	Primary storage	Secondary storage
Example	RAM	ROM
Scenario - Data retention 数据保留	Data lost when power removed	Data retained when power removed
Scenario - Data loss due to error	Data lost on power crash	Data usually retained

2. Memory Enhancement - 3 key approaches（Page 6 -15）
Wider path memory access
- Several bytes can be read or written between CPU & memory with each access

Memory interleaving
- Divide memory into, for multiple access at the same time.
- Each part has its own MAR & MDR.

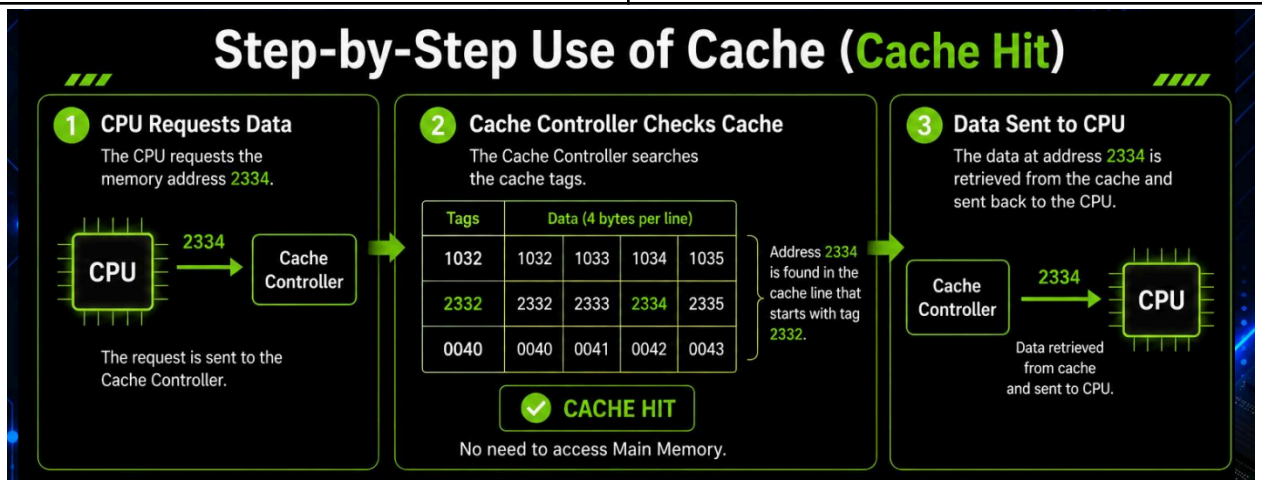
Cache Memory
在**3.**那边

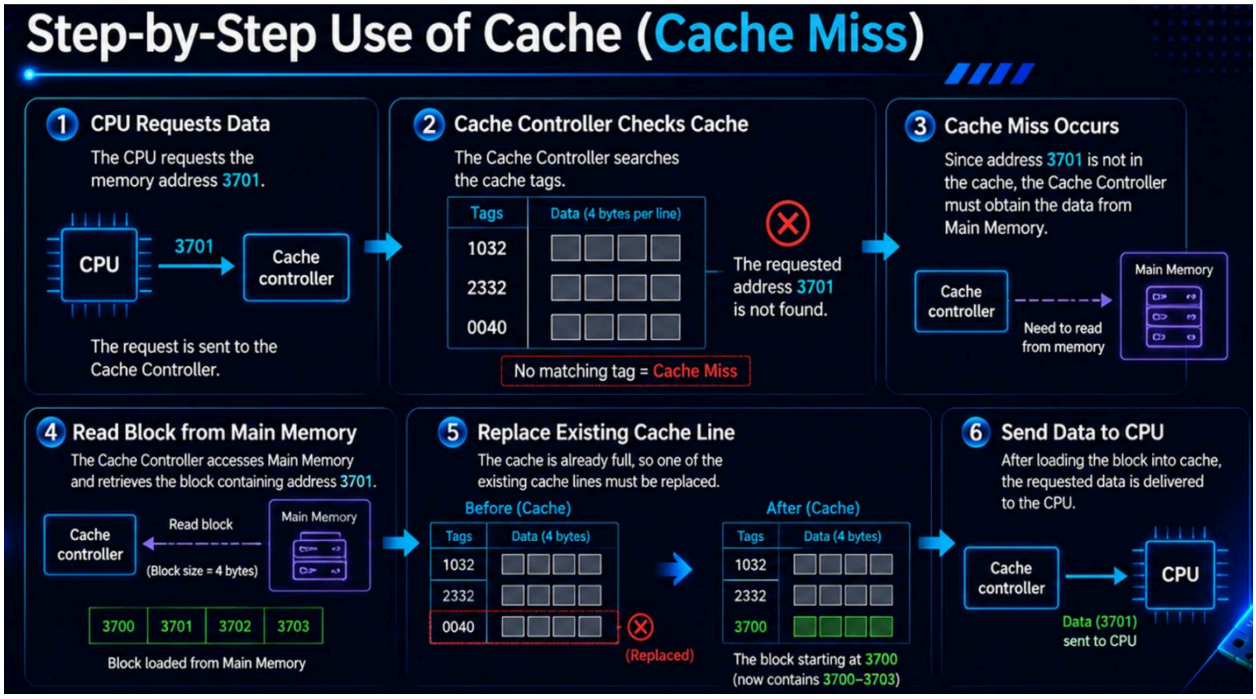
3. Purpose and Performance Benefits of Caching memory / Caching memory concepts (Page 10-15)
- Small amount of fast memory

- Sits between normal main memory and CPU

4. Cache Hit VS Cache Miss

Step by step use of (题目可能会要求画diagram)	
Cache Hit	Cache Miss
1. CPU requests data 2. Cache Controller Checks Cache 3. Data Sent to CPU	1. CPU requests data 2. Cache Controller Checks Cache 3. Cache Miss Occurs 4. Read Block from Main Memory 5. Replace Existing Cache Line 6. Data Sent to CPU





Compare	
Cache hit	Cache Miss
requested data is found in the cache	requested data is not found in the cache
Very fast	slower

Extra:
Describe what will happen when a cache memory is full.
When the cache is full, one of the existing cache lines must be replaced.

5. Paging - Key Concepts (Virtual Memory Management) (Page 17-21)

- Main memory and virtual memory are divided into equal-sized pages.
- A process does not need to reside entirely in memory at one time.
- Uses both RAM and secondary storage (disk).

6. TLB (Translation Lookaside Buffer) (Pages 31)
- A special fast lookup cache which is associative.

7. The need of TLB / Advantages (Pages 32)
Faster Memory Access
- Reduces the time to fetch data from main memory.

Improves System Performance
- Reduces effective access time (EAT).

Avoids Double Memory Access

8. Limitations of TLB / Disadvantages (Page 35)
Limited Size
- TLBs have a finite number of entries.

Complex Hardware
- TLBs require additional hardware components.

Management overhead

Chap 4 calculation part (will come out question at Q3 OR Q4)

1. Paging Address Translation (Chap 4, P27–29)

Example question:

Logical address size = 20 bits
Page size = 4 KB
Logical address = 95A52H
Page table entry:

Page No	Frame No
:	:
95H	3A7H
:	:

(i) Determine the numbers of bits used for page number and offset.
Logical address = Page No + Offset
4 KB = 4096 bytes = 2^{12} bytes
Offset bits (d) = 12 bits
Page number bits (p) = 20 − 12 = 8 bits
Page No 8 bit / Offset 12 bit
95 = Page no / A52 = Offset

(ii) Determine the frame number.
看回去page table entry
Page 95 → Frame 3A7
The frame no is 3A7H

(iii) Construct the physical address.
= Frame Number + Offset
= 3A7 + A52
Therefore : Physical Address = 3A7A52H

2. EAT (Effective Access Time) (Page 33-34)

📖 教学：EAT 的核心概念

两种情况

情况	发生概率	需要的时间
TLB Hit	α (Hit Rate)	TLB Access + Memory Access
TLB Miss	(1 - α)	TLB Access + 2 × Memory Access

为什么 TLB Miss 要 2 次 Memory Access?

- 第 1 次：去 Page Table 找 Frame Number
- 第 2 次：用 Frame Number + Offset 去实际内存取数据

Format: EAT = (TLB Access + Memory Access) × α
+ (TLB Access + 2 × Memory Access) × (1 - α)

Paper Question:

d) Effective Access Time (EAT) is the average time required to access a memory location, taking into account the probabilities of Translation Lookaside Buffer (TLB) hits and misses, along with their respective access times.

Given that:

- TLB Access Time: 10 ns
- Memory Access Time: 80 ns
- TLB Hit Rate: 90%

Calculate the EAT and present the final result in nanoseconds (ns). (5 marks)

TLB Access = 10 ns
Memory Access = 80 ns
TLB Hit: α = 0.90
TLB Miss: 1 - α = 0.10

Calculation
EAT = (10 + 80) × 0.90 + (10 + 2×80) × 0.10

= (90) × 0.90 + (10 + 160) × 0.10
= 81 + (170) × 0.10
= 81 + 17
= 98 ns

Ans: 98 ns

3. Segmentation Physical Address (Page 36-38)

Paper Question:

b) Given the following segment and offset addresses:

- Stack Segment (SS) = 0x3000
- Stack Pointer (SP) = 0x0050
- Base Pointer (BP) = 0x0020
- Code Segment (CS) = 0x2000
- Instruction Pointer (IP) = 0x0060
- Data Segment (DS) = 0x4000
- Extra Segment (ES) = 0x5000

(i) Calculate the absolute address of the next instruction to be executed. (4 marks)

(ii) Calculate the absolute address of the top (most recent) item in the stack segment. (4 marks)

📖 第一步：理解题目在问什么

题目给了很多 segment 和 offset，但你要知道 **哪个 segment 配哪个 offset**。

配对规则（一定要记住）

用途	Segment	Offset
Next instruction	CS (Code Segment)	IP (Instruction Pointer)
Top of stack	SS (Stack Segment)	SP (Stack Pointer)
Data access	DS (Data Segment)	题目给的 offset
Extra data	ES (Extra Segment)	题目给的 offset
Stack frame (local variables)	SS (Stack Segment)	BP (Base Pointer)

所以：

- (i) Next instruction → **CS + IP**
- (ii) Top of stack → **SS + SP**

> ⚠️ 注意：题目给了 BP，但 (ii) 问的是 **top (most recent) item**，所以用 **SP**，不是 BP。
> BP 是用来访问 stack frame 里的 local variables 的。

(i)
第二步：写出format:
Physical Address = (Segment × 16) + Offset
为什么 × 16？
Segment register 是 16 bits
Physical address 是 20 bits
所以 segment 要左移 4 bits（等于 × 16），再加上 offset

计算 (i) Next Instruction
找配对 - Next Instruction -> CS + IP：
CS = 0x2000
IP = 0x0060

Physical Address = (CS × 16) + IP
= (0x2000 × 0x10) + 0x0060
= 0x20000 + 0x0060
= 0x20060

Ans: Physical Address = 0x20060

(ii)
计算 (ii) Top of Stack

找配对 - Next Instruction -> SS + SP：
SS = 0x3000
SP = 0x0050

Physical Address = (SS × 16) + SP
= (0x3000 × 0x10) + 0x0050
= 0x30000 + 0x0050
= 0x30050

Ans: Physical Address = 0x30050